

Supermarket Sales Data Analysis Report

1. Data Overview

The dataset used in this analysis is the “Supermarket Sale” dataset. The dataset contains information about supermarket transactions, including customer type, product category, quantity sold, and total sales amount. The dataset contains approximately 254 rows and 8 columns. This dataset is useful for analysing sales performance and customer purchasing behaviour.

2. Data Cleaning

During the data cleaning process, missing values in the “Quantity” column were replaced using the average value of the column, which was 11. For missing values in text columns such as “Customer_type” and “Product_category”, mode was used to replace the blank cells. In addition, duplicate rows were identified and removed to improve data accuracy and data quality.

3. Descriptive Statistics and Visualisation

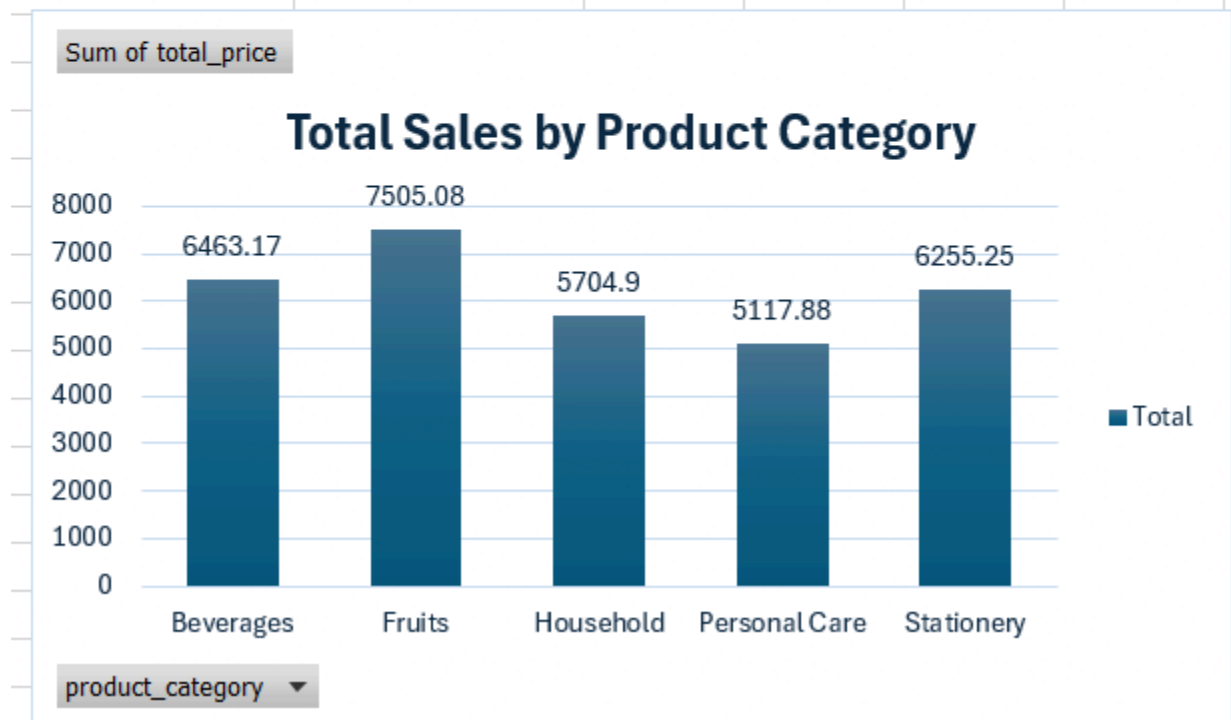
Several visualisations were created to better understand the dataset, including:

- Total Sales by Product Category
- Sales by City
- Customer Type Distribution

Based on descriptive statistics, the average sales amount was approximately 124.19 USD. The highest sales value recorded was 427.14 USD, while the minimum sales value was 2.18 USD.

3.1. Sales by Product Category

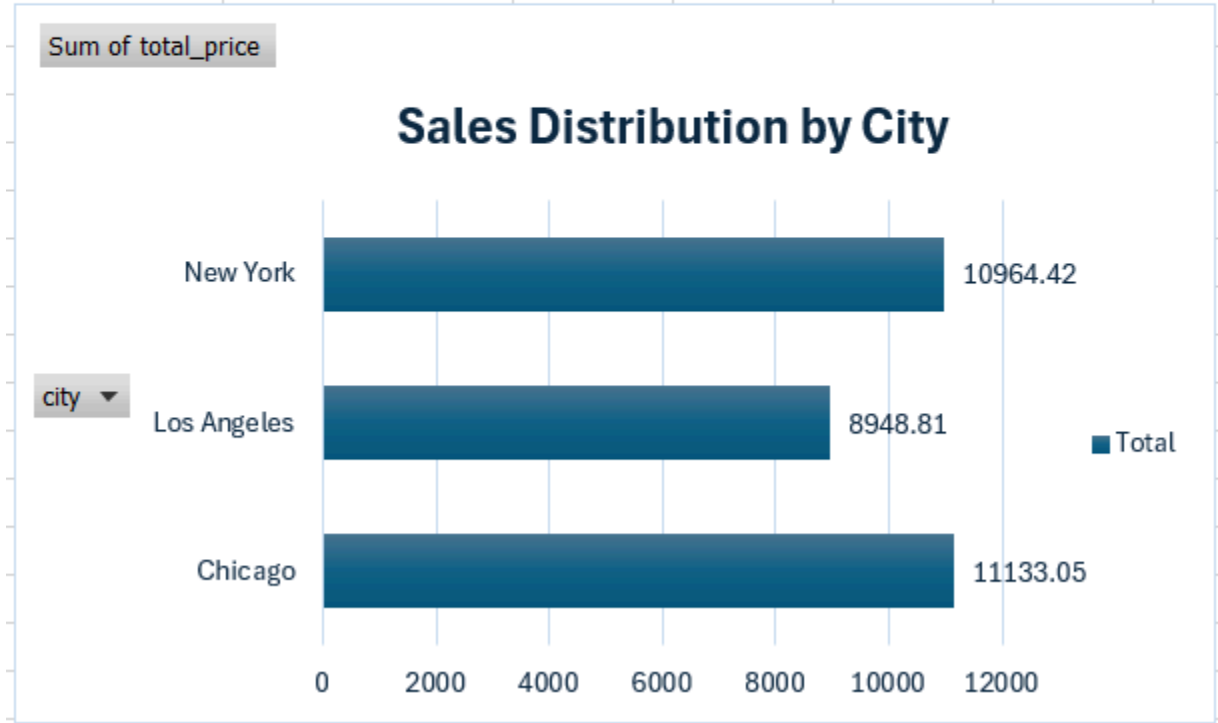
Row Labels	Sum of total_price				
Beverages	6463.17				
Fruits	7505.08				
Household	5704.9				
Personal Care	5117.88				
Stationery	6255.25				
Grand Total	31046.28				



Insight 1: The Fruits category generated the highest total sales among all product categories. This suggests that fruit products are the most demanded items and contribute significantly to the supermarket's overall revenue.

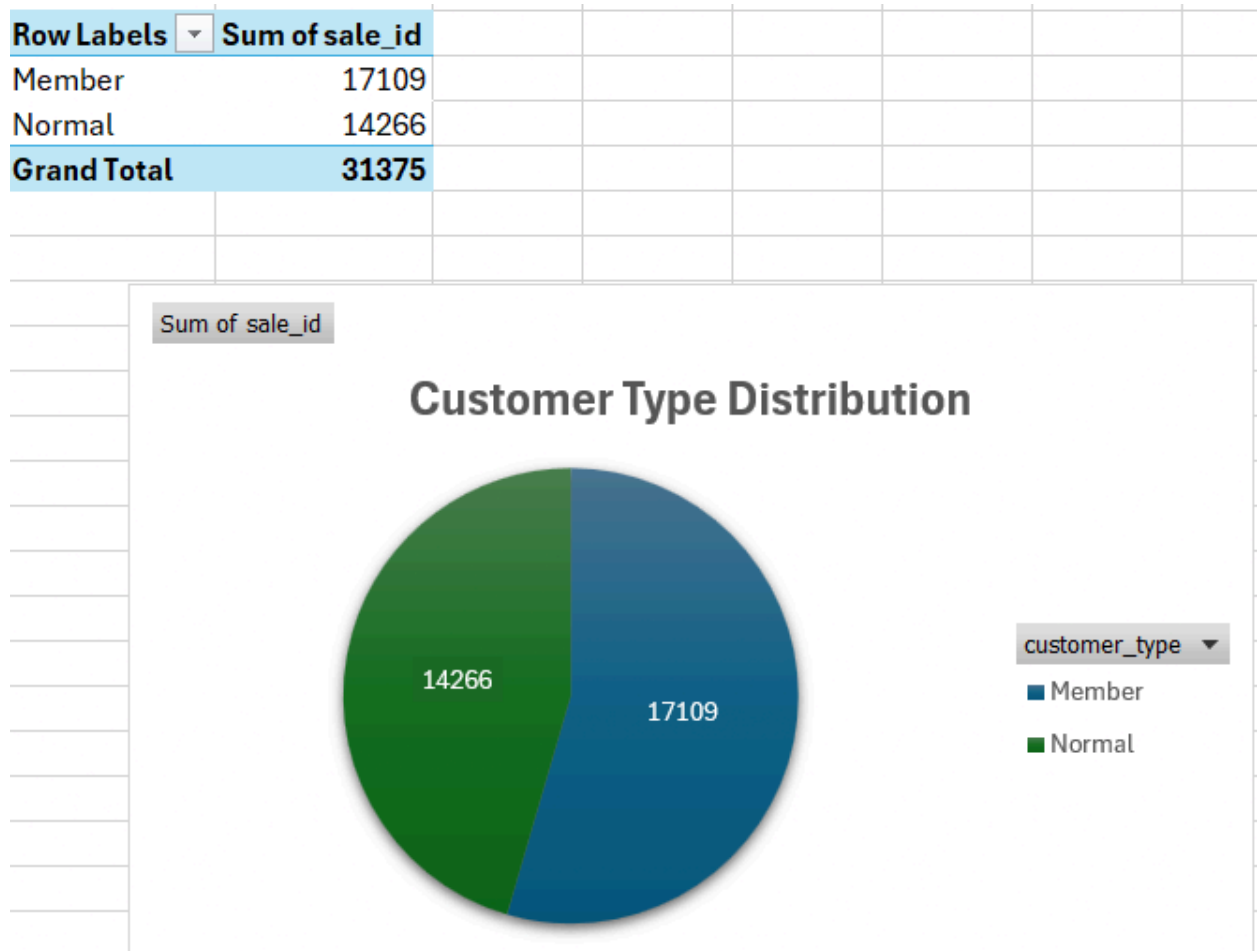
3.2. Sales Distribution by City

Row Labels	Sum of total_price
Chicago	11133.05
Los Angeles	8948.81
New York	10964.42
Grand Total	31046.28



Sales performance differed across cities, indicating variations in customer purchasing activity.

3.3. Customer Type Distribution



Insight 2: Member customers made more purchases compared to normal customers. This indicates that loyalty membership programs may encourage customers to shop more frequently and increase customer retention.